Supplementary Evidence for MPC Interface Codes

World Models for Autonomous Driving: From Future Generation to Decision Making

This supplement audits the 30 method rows in manuscript Tables II–IV: 29 temporal world models and RangeLDM as a static sensor-generation boundary comparator. Profiles and bracketed reference numbers match the revised manuscript. These are the survey authors' evidence annotations, not capability scores reported by the original papers.

Reading the codes. A denotes action-conditioned rollout; C, cost/reward exposure; H, constraint handling; U, decision-usable uncertainty; and R, replanning or executable agent–world feedback. E requires an implemented interface or a reported evaluation within the stated scope. P denotes partial or indirect support. – means that the audited source does not provide sufficient evidence, rather than asserting that the capability is impossible.

Scope of evidence. A distinguishes a supplied action/trajectory or action-dependent latent transition from joint prediction without a distinct intervention. C distinguishes an exposed cost/reward or selection score from a generic training objective or an evaluation metric alone. H=E can refer to soft penalties, filtering, or non-ego handling; it does not guarantee ego safety. U=P includes stochastic forecasts, latent distributions and mode scores. U=E requires predictive uncertainty to change a decision rule, as in MP3's map-uncertainty cost; classifier confidence or sampling alone is insufficient. R distinguishes policy-only execution (P) from repeated rollout-based planning or an implemented agent–simulator feedback loop (E). ReSim's model-only visual loop is explicitly separated from traffic/kinematic simulation and is not evidence of validated physical response fidelity.

Source anchors. Links identify the audited primary source/version. PDF-page numbers refer to physical PDF pages; proceedings page numbers are used for DrivingGPT. The supplied preprint versions are retained where specified. DrivingGPT's next-token objective was also checked against the ICCV 2025 paper. Each row explains all five marks, including every E. No performance result is recomputed or added.

Table S1: Future generation and the static sensor-generation boundary

Study / primary source	A/C/H/U/R	Per-entry evidence and limits
FIERY [33] 2104.10490v3	`--/--/--/P/--`	**A=–:** Past ego-motion aligns observations; future candidate ego actions are not queried. **C=–:** No planner-facing cost or reward module is implemented. **H=–:** The reported system forecasts instances rather than enforcing ego feasibility. **U=P:** Gaussian future sampling represents multimodality, without uncertainty-dependent control. **R=–:** Evaluation concerns logged future prediction, not executed replanning. *Anchors:* Secs. 3.3–3.5 and 4; PDF pp. 3–5.
BEVWorld [34] 2407.05679v3	`E/--/--/P/--`	**A=E:** Ego-motion tokens modulate future BEV diffusion, explicitly conditioning the forecast on movement and steering (Sec. 3.2, Eq. 6). **C=–:** Sensor prediction and transfer tests do not implement a cost/reward head. **H=–:** Ray-based reconstruction supplies sensor geometry, not action-feasibility handling. **U=P:** The stochastic future diffusion distribution is available, without a risk-dependent decision rule. **R=–:** The reported evaluation forecasts sensors; it does not repeatedly execute model-based decisions. *Anchors:* Sec. 3.2; Secs. 4.2–4.3; PDF pp. 5–10.
GAIA-1 [35] 2309.17080v1	`E/--/P/P/--`	**A=E:** Action embeddings condition token prediction; speed and steering controls are demonstrated (Secs. 2.1, 2.3 and 5). **C=–:** No candidate cost or reward evaluator is implemented. **H=P:** Action/text conditioning provides indirect control, without a feasibility filter. **U=P:** Stochastic conditional video futures do not supply an uncertainty-dependent driving rule. **R=–:** Autoregressive video extension is not execution followed by new environmental feedback. *Anchors:* Secs. 2.1, 2.3 and 5; PDF pp. 4–5 and 10–16.
DriveDreamer [36] 2309.09777v2	`E/--/P/P/--`	**A=E:** ActionFormer uses ego driving actions in latent transitions and predicts future structural conditions (Sec. 3.2, Eqs. 5–8). **C=–:** Generating future actions does not expose a separate planner cost/reward. **H=P:** HD-map and box conditioning supplies structural information without a deployed feasibility operator. **U=P:** Gaussian ActionFormer states and diffusion futures are stochastic, without risk-aware decision use. **R=–:** The reported action/video tests do not establish repeated executed replanning. *Anchors:* Secs. 3.1–3.2 and 4.2–4.3; PDF pp. 4–8.

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Table S1 continued

Study / primary source	A/C/H/U/R	Per-entry evidence and limits
DriveDreamer-2 [37] AAAI 2025	P/--/P/P/--	**A=P:** Prompt-derived trajectories/layouts control videos indirectly; an online candidate-action transition interface is not established. **C=–:** No planner cost/reward head is reported. **H=P:** Trajectory and map generation impose scene conditions, without executed feasibility enforcement. **U=P:** Diffusion represents a generative future distribution, without decision-usable risk estimation. **R=–:** Video generation and augmentation tests do not execute receding-horizon decisions. *Anchors:* Method: trajectory generator, HDMap generator and UniMVM; PDF pp. 3–5.
Copilot4D [38] 2311.01017v4	E/--/--/P/--	**A=E:** The predictive distribution explicitly conditions on ego SE(3) poses as actions (Sec. 4, Eq. 3; Sec. 4.3). **C=–:** Point-cloud forecasting has no planner cost/reward head. **H=–:** The tokenizer and ray renderer do not implement action-feasibility handling. **U=P:** Multiple plausible point-cloud futures are shown, but no margin, risk objective or action rule uses their uncertainty (Fig. 1). **R=–:** Forecast rollout uses generated tokens rather than environmental execution feedback. *Anchors:* Sec. 4, Eq. 3; Sec. 4.3; Fig. 1; PDF pp. 1, 4–6.
OccWorld [39] 2311.16038v1	P/--/P/P/--	**A=P:** Ego and occupancy tokens evolve jointly; supplied candidate ego interventions are not separately evaluated. **C=–:** The ego displacement loss/output is not a planner-facing cost module. **H=P:** Semantic occupancy is an indirect constraint representation; no executed feasibility module is shown. **U=P:** Categorical world-token distributions supply indirect uncertainty representation only. **R=–:** nuScenes forecasting/planning is evaluated on logged futures. *Anchors:* Secs. 3.3–3.4 and 4; PDF pp. 4–7.
DOME [40] 2410.10429v1	E/--/P/P/--	**A=E:** Relative ego translation/yaw conditions adaLN, and trajectory resampling tests control (Sec. 3.3, Fig. 3). **C=–:** The reported generator has no planner cost/reward module. **H=P:** Map-based feasible trajectory resampling is a training prior, not executed constraint enforcement. **U=P:** Conditional stochastic occupancy futures are not used by an uncertainty-aware decision rule. **R=–:** Repeated occupancy sampling extends prediction without executing a driving policy. *Anchors:* Secs. 3.2–3.4 and 4; PDF pp. 5–10.
RangeLDM [41] 2403.10094v2 *Static comparator*	--/--/--/--/--	**A=–:** This is a single-scan sensor generator, not a candidate-conditioned temporal transition. **C=–:** No driving decision cost/reward interface is reported. **H=–:** Range projection/rendering is not driving-feasibility handling. **U=–:** Static sample diversity does not represent uncertainty over a scene-conditioned temporal future. **R=–:** Generation, upsampling and inpainting do not provide an executed driving feedback loop. *Anchors:* Fig. 1; Secs. 4.1–4.3 and 5; PDF pp. 1, 5–10.
SparseWorld [42] 2510.17482v1	P/--/P/--/--	**A=P:** Ego-state queries condition forecasting; a supplied candidate-action intervention interface is not established. **C=–:** Joint trajectory regression is not an exposed cost/reward evaluator. **H=P:** Occupancy and spatially modulated queries provide indirect geometry for planning. **U=–:** The reported query-regression rollout does not expose a future distribution or risk measure. **R=–:** The reported nuScenes tests do not execute receding-horizon decisions. *Anchors:* Method: State-Conditioned Continuous Forecasting, Eqs. 6–7; PDF pp. 4–7.

Table S2: Planning with world models

Study / primary source	A/C/H/U/R	Per-entry evidence and limits
MP3 [67] 2101.06806v1	P/E/E/E/E	**A=P:** Candidate trajectories query shared scene forecasts, which are not themselves conditioned on candidate ego actions. **C=E:** Explicit map, route, collision, comfort and progress terms score trajectories (Sec. 3.3.3). **H=E:** Road and occupancy penalties directly enter trajectory selection; these are soft handling, not hard guarantees (Sec. 3.3.3). **U=E:** Map-distribution standard deviation is multiplied by ego speed in a cost, promoting slower motion under uncertainty (Sec. 3.3.3). **R=E:** The planner is evaluated after its own executed motion changes the observed state (Sec. 4, Table 1). *Anchors:* Secs. 3.3.3 and 4, Table 1; PDF pp. 6–7.
MILE [50] 2210.07729v2	E/--/P/P/P	**A=E:** The prior Gaussian explicitly conditions on the policy-predicted previous action; the posterior uses the expert action (Sec. 3.4). This is an action-conditioned latent transition, not an exposed candidate-search API. **C=–:** The paper explicitly trains without rewards; the driving policy is learned by imitation. **H=P:** BEV supervision and learned behavior are indirect, rather than a deployed feasibility operator. **U=P:** Stochastic latent futures can be decoded into multiple BEV predictions (Sec. 5.4). **R=P:** CARLA recurrent-policy execution and observing/imagining tests are reported; online candidate-cost reoptimization is not (Secs. 5.3, 6.2). *Anchors:* Secs. 3.1, 3.4–3.5, 5.3–5.4 and 6.2; PDF pp. 3, 6, 8–10.
DTPP [68] 2310.05885v2	E/E/E/--/E	**A=E:** Ego trajectory-tree branches condition predictions of other agents (Sec. III-B; Alg. 1). **C=E:** A context-weighted cost over ego, learned interaction and collision features scores branches (Sec. III-C, Eq. 2). **H=E:** The planner drops dynamically invalid trajectories and includes collision costs (Secs. III-A, III-C). **U=–:** Ego plan branches and cost-softmax training do not establish predictive uncertainty over agent outcomes. **R=E:** nuPlan reactive and non-reactive closed-loop planning are reported (Sec. IV-A, Tables I–II). *Anchors:* Secs. III-A–III-C and IV-A–IV-C; PDF pp. 3–6.
Think2Drive [69] 2402.16720v2	E/E/P/P/P	**A=E:** The RSSM advances latent states conditioned on ego actions (Sec. 3.2, Eq. 2). **C=E:** Reward/termination predictors and imagined returns train the actor–critic (Sec. 3.2). **H=P:** Safety-related rewards, terminations and steering shaping are indirect deployment constraints. **U=P:** Stochastic latent dynamics represent futures, without explicit uncertainty-dependent control. **R=P:** CARLA executes the learned policy; imagination primarily supports training, not online candidate optimization. *Anchors:* Sec. 3.2, Eq. 2; Secs. 3.3 and 4; PDF pp. 5–10.
Drive-OccWorld [60] AAAI 2025	E/E/E/P/--	**A=E:** Velocity, steering, trajectory and command conditions control future occupancy/flow (Sec. 3.3). **C=E:** An explicit occupancy-based cost selects among trajectory proposals (Sec. 3.4). **H=E:** Agent, road and volume geometry directly enters cost terms; cost ablations test their effects (Sec. 3.4; Table 8). **U=P:** Occupancy prediction supplies indirect state uncertainty; no uncertainty-dependent margin or fallback is demonstrated. **R=–:** The reported nuScenes planning tests are open loop. *Anchors:* Secs. 3.3–3.4; Secs. 4.3–4.4, Table 8; PDF pp. 4–7.
Drive-WM [70] CVPR 2024	E/E/E/P/--	**A=E:** Proposed maneuvers condition generated multiview futures (Sec. 4.1). **C=E:** Perception on generated images supplies object/map rewards for trajectory selection (Sec. 4.2). **H=E:** Curb, centerline and object-distance penalties explicitly shape selection; they remain perception-dependent soft handling (Sec. 4.2). **U=P:** Stochastic visual futures are not propagated as a calibrated risk or uncertainty-aware control rule. **R=–:** A planning tree advances in imagination; this is not execution with fresh environmental feedback. *Anchors:* Secs. 4.1–4.2, Fig. 5, Table 4; PDF pp. 5–7.

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Table S2 continued

Study / primary source	A/C/H/U/R	Per-entry evidence and limits
AdaptiveDriver [71] 2406.10714v3	E/E/E/P/E	**A=E:** Candidate ego proposals are simulated with reactive agents controlled by learned IDM parameters (Sec. III, Fig. 5). **C=E:** The PDM-derived proposal-scoring interface selects the highest-scoring simulated trajectory (Fig. 5). **H=E:** IDM dynamics and PDM-style road/collision-aware proposal scores implement scoped feasibility handling (Fig. 5; PDM-C Sec. 3.3 [92]). **U=P:** Behavior-cluster classification provides indirect behavioral uncertainty, without calibrated risk propagation. **R=E:** The MPC planner executes selected proposals in reactive nuPlan evaluation (Sec. IV, Tables II–V). *Anchors:* Sec. III, Figs. 3 and 5; Sec. IV; PDF pp. 3–5. Inherited scoring: PDM-C Sec. 3.3.
WoTE [72] 2504.01941v2	E/E/P/P/E	**A=E:** Trajectory proposals are encoded to query corresponding future BEV states (Sec. 3.2). **C=E:** Future-aware imitation/simulation rewards rank candidates (Sec. 3.3, Eq. 4). **H=P:** Learned collision/drivable-area reward predictions are soft surrogates, not explicit geometric feasibility checks. **U=P:** Probabilistic reward heads are indirect outcome confidence; uncertainty is not separately propagated or calibrated. **R=E:** Online trajectory evaluation is executed in the reactive Bench2Drive test (Table 2). *Anchors:* Secs. 3.2–3.4, Eq. 4; Table 2; PDF pp. 3–6.
Raw2Drive [73] 2505.16394v2	E/E/P/P/P	**A=E:** Aligned raw/privileged RSSMs advance latent states conditioned on actions (Secs. 3.1–3.2). **C=E:** Privileged reward/continuation heads supply returns during raw-policy imagination (Sec. 3.2, Fig. 5). **H=P:** Reward-based safety learning is indirect; a deployed trajectory-feasibility operator is not reported. **U=P:** Stochastic RSSM states provide indirect future uncertainty, without uncertainty-aware decision validation. **R=P:** CARLA executes the trained policy; the aligned world-model rollout mainly supports training. *Anchors:* Secs. 3.1–3.2, Figs. 4–5; Sec. 4; PDF pp. 4–8.
AdaWM [74] 2501.13072v2	E/E/P/P/P	**A=E:** The learned recurrent transition explicitly conditions on actions (Sec. 2, Eq. 1). **C=E:** Predicted reward and a terminal value train the policy through imagined returns (Sec. 2, Eq. 2). **H=P:** Safety rewards in Appendix C are indirect, not an enforced runtime constraint set. **U=P:** The stochastic RSSM supports partial U; TV mismatch used to select finetuning is not predictive-risk calibration. **R=P:** CARLA interaction/adaptation executes a learned policy; online candidate-trajectory reoptimization is not established. *Anchors:* Sec. 2, Eqs. 1–2, Alg. 1; Apps. C and E; PDF pp. 4–7, 17, 19–20.

Table S3: Hybrid prediction–planning and interactive simulation

Study / primary source	A/C/H/U/R	Per-entry evidence and limits
GenAD [95] 2402.11502v3	P/P/P/P/--	**A=P:** Ego and other-agent futures share latent generation; supplied ego interventions are not independently evaluated. **C=P:** Driving-specific trajectory/constraint losses support an indirect cost connection, not a runtime cost head. **H=P:** Collision, boundary and lane-direction losses shape training rather than enforce executed feasibility. **U=P:** A Gaussian trajectory latent explicitly models alternative futures, without a risk-dependent decision rule. **R=–:** The reported nuScenes planning evaluation is open loop. *Anchors:* Secs. 3.2–3.4 and 4; PDF pp. 4–7.
DrivingGPT [96] ICCV 2025	E/--/P/P/--	**A=E:** Action tokens condition visual-token prediction in the unified autoregressive sequence (Sec. 3.2, Eq. 1). **C=–:** The implemented objective is next-token cross-entropy; the generic MDP reward definition is not an implemented cost/reward head. **H=P:** Empirical action-bin clipping is a limited action-range prior, not geometric feasibility enforcement. **U=P:** Autoregressive future-token distributions support stochastic alternatives, not uncertainty-dependent driving. **R=–:** The reported planning evaluation uses NAVSIM; no executed agent–world feedback loop is demonstrated. *Anchors:* Sec. 3.2, Eq. 1; Sec. 4.1 and planning evaluation. ICCV pp. 26892–26896.
World4Drive [97] 2507.00603v1	E/E/P/P/--	**A=E:** Intention-aware trajectory encodings condition future latent generation (Sec. 3.3.1). **C=E:** ScoreNet ranks future latents and selects a trajectory at inference (Sec. 3.3.2, Eq. 7). **H=P:** Scene semantics/geometry inform learned selection, without an explicit executed feasibility operator. **U=P:** Intention-mode scores are not a calibrated physical uncertainty measure or an uncertainty-dependent safety rule. **R=–:** nuScenes/NAVSIM tests do not demonstrate reactive repeated execution. *Anchors:* Secs. 3.3.1–3.3.2, Eq. 7; Sec. 4.1; PDF pp. 4–5.
DriveArena [98] 2408.00415v1	E/P/P/P/E	**A=E:** Agent trajectories update ego motion in Traffic Manager before the next generated observation (Sec. 3.4). **C=P:** PDMS/ADS are reported simulator-evaluation outputs; a built-in candidate-rollout optimizer or reward learner is not shown. **H=P:** Collision detection and scene conditions are not an explicit planner feasibility-enforcement interface. **U=P:** Stochastic World Dreamer/traffic generation is not uncertainty-aware ego control. **R=E:** Traffic Manager executes the ego trajectory and feeds the next rendered state to the agent (Sec. 3.4). *Anchors:* Secs. 3.1–3.4, Eqs. 1–3; PDF pp. 4–6.
DriveLaW [99] 2512.23421v2	P/--/P/P/--	**A=P:** Kinematics are encoded as text and video latents condition actions; a tested numeric candidate-action rollout interface is not established. **C=–:** Flow-matching/imitation trains the action decoder; a planner cost/reward head is not implemented. **H=P:** Shared video latents supply indirect scene geometry rather than an explicit feasibility operator. **U=P:** Video/action diffusion represents stochastic alternatives, without decision-usable risk estimates. **R=–:** Reported NAVSIM planning does not demonstrate reactive execution. *Anchors:* Secs. 3.2–3.4 and 4; PDF pp. 4–7.
UniDrive-WM [100] 2601.04453v3	E/P/P/P/P	**A=E:** Future image tokens explicitly condition on the planning token (Sec. 3.3, Eq. 7). **C=P:** Collision/boundary training losses support an indirect cost connection; there is no separate rollout-to-cost head. **H=P:** Training-time collision and boundary losses do not implement runtime feasibility enforcement. **U=P:** The planner has a Gaussian trajectory latent, without an uncertainty-dependent control rule. **R=P:** Bench2Drive reports reactive learned-policy execution, without online candidate-world-model optimization. *Anchors:* Secs. 3.2–3.4, Eq. 7; Sec. 4, Table 1; PDF pp. 4–7.

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Table S3 continued

Study / primary source	A/C/H/U/R	Per-entry evidence and limits
WorldDrive [101] 2603.14948v1	E/E/P/P/--	**A=E:** Trajectory encodings condition future scene-latent generation in TA-DWM (Sec. 3.1). **C=E:** A distilled future-aware rewarder scores candidates and selects the highest reward (Secs. 3.3–3.4, Eq. 7). **H=P:** Learned simulation/preference scores are indirect feasibility surrogates. **U=P:** Multimodal generation/planning supplies alternatives without uncertainty-aware risk propagation. **R=–:** The reported nuScenes/NAVSIM tests do not demonstrate reactive execution. *Anchors:* Secs. 3.1–3.4, Figs. 2–3, Eq. 7; Sec. 4; PDF pp. 3–6.
DriveWorld-VLA [102] 2602.06521v1	E/E/P/P/--	**A=E:** A future action-conditioned flow branch generates BEV latents (Sec. 3.2, Eqs. 8–9). **C=E:** A learned scalar reward evaluates imagined outcomes and weights action refinement (Sec. 3.3, Eqs. 10–12). **H=P:** Reward/semantic supervision is indirect, without a runtime feasibility operator. **U=P:** Generative BEV futures represent partial uncertainty; no uncertainty-dependent decision rule is tested. **R=–:** Reward-guided training and NAVSIM outcomes are not reactive repeated execution. *Anchors:* Secs. 3.2–3.3, Eqs. 8–12; Sec. 4; PDF pp. 4–6.
VectorWorld [55] 2603.17652v1	E/P/E/P/E	**A=E:** Executed ego trajectories enter the world-state step and streaming condition update (App. A.12.3, Alg. 2). **C=P:** NPC return conditioning/kinematic costs do not expose a complete ego candidate-cost interface. **H=E:** NPC kinematic costs shape action logits, with alignment loss and physics-violation tests (Sec. 3.6, Eqs. 9–10). This scope is non-ego. **U=P:** Stochastic graph completion and NPC actions are not an uncertainty-aware ego risk rule. **R=E:** The loop re-observes, replans and steps ego/NPC state before outpainting (Alg. 2; Table 2). *Anchors:* Sec. 3.6, Eqs. 9–10; Table 2; App. A.12.3, Alg. 2.
ReSim [76] 2506.09981v1	E/E/P/P/E	**A=E:** Ego waypoint sequences condition future video generation (Sec. 2.2). **C=E:** Video2Reward scores action-conditioned futures for policy selection (Secs. 2.3–2.4). **H=P:** A learned scalar infraction reward is an indirect feasibility surrogate. **U=P:** Stochastic visual forecasts are not propagated as calibrated risk or uncertainty-dependent margins. **R=E:** Actions generate the next video, which is fed back to the policy (Secs. 2.4, 3.2; Fig. 8). This is a model-only visual loop demonstrated from two seeds. *Anchors:* Secs. 2.2–2.4 and 3.2; Fig. 8; PDF pp. 3–6 and 8.